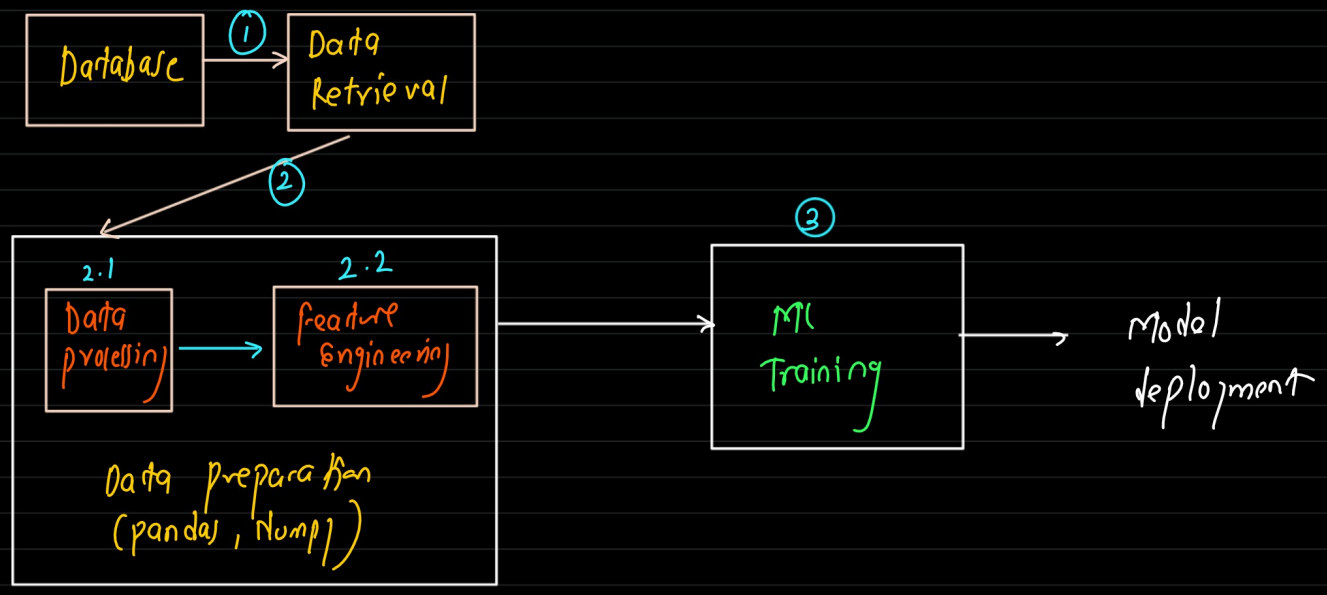
22-April-2026

Feature Engineering

numpy, pandas, data visualization, statistics

process of using domain knowledge to select, transform & create new features from raw data → make machine learning model more accurate & effective



Feature Engineering :—

→ Handle Missing data

→ Handling imbalance data

→ Handling outliers

→ Feature Scaling

→ Data Encoding

→ Feature Selection

Feature Engineering

80 columns

80% →

10% →

Machine Learning

X

[a, b, c, d, e, f]

→ independent

y ← we want to predict

sale price

→ dependent

A, B, C, D, E, F

sales_price

lower_core

-ve ⟷ +ve

-0.1 0.1

0.1 → filter

FE → ML

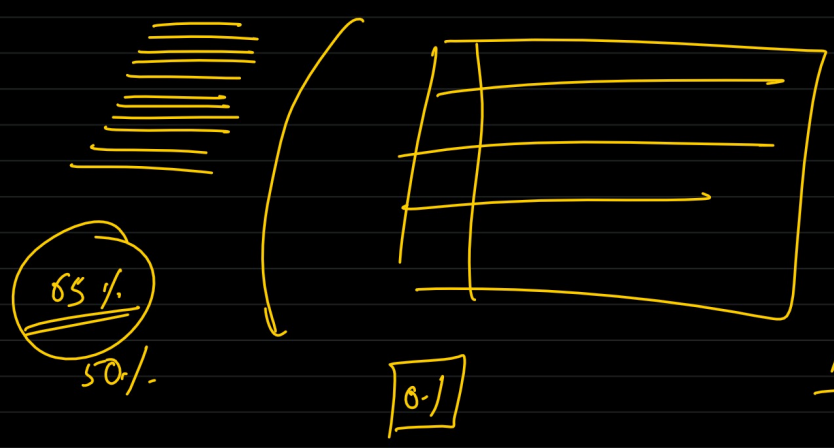
X → feature column

Y → predict

a → y → 0.1
b → y → 0.11
c → y → 0.9

a, b, c → y

a → y → 0 0.1 1
b → y
c → y



heatmap - DJ

heatmap - 88

Multiplication table

A	B
1	2
2	4
3	6
4	8
5	10
6	12
7	14
8	16
9	18
10	20

→ common step size

A → increasing
→ step-size → 1

B → increasing
→ step-size → 2
→ common step size

A^{*2} → B
↓
A *2

correlated ← A B → high correlation

A	B
1	2
2	4
3	12
4	84

→ stash → low core

x - feature

A B C D E

(sale)

multi-collinearity

↓

comparing correlation within features → independent variable

feature

R → red	→ red
G → green	→ green
B → Blue	→ blue

orange

fruit

A	B
	proportion
1	100
2	200
3	300

→ high

A B C D E Salepric

A — 1.0 — salepric

ML

$2 \times 1 = 2$
$2 \times 2 = 4$
$2 \times 3 = 6$

$3 \times 1 = 3$
$3 \times 2 = 6$
$3 \times 3 = 9$

1, 2
2, 4
3, 4
1, 3
2, 6
3, 9

↓
ML → find the for

2
2
2
3
3
3

2 → ML → 2
3 → ML → 3

X — ML — Y

A, B, C, D — sale pric

A → ML → A

A, B, C, D, Sale_pric → redunt columns